Preliminary Bill of Materials: Autonomous Magnetic Chessboard

ECE445L Fall 2026, Lab 7A — Prices are rough estimates; generated from `BOM.csv`

1a = cost of all components, wherever they come from. **1b** = cost of components that count toward the competition limit (excludes checkout desk, professor's parts, free samples, parts every group already owns, TIW resources, and the PCB). Power and enclosure (Lab 7B) and the stretch-goal SPI flash are listed but not totaled.

Ref	Item	Example part	Qty	Unit $	Ext $	1b?	Source	Lab
U1	Microcontroller	M0G3507QDGS32RQ1 (VSSOP-32)	1	1.50	1.50	No	Professor (with Lab 10 PCB)	10
-	Development board	LP-MSPM0G3507 LaunchPad	1	15.00	15.00	No	Already owned	7A
M1-M2	Stepper motor	NEMA 17 bipolar 1.5 A 42 N-cm	2	10.00	20.00	Yes	Amazon (2-pack)	7A
U2-U3	Stepper driver module	TI DRV8825 carrier (Pololu-style)	2	2.50	5.00	Yes	Amazon (5-pack)	7A
C1-C2	Driver bulk capacitor	100 uF 25 V electrolytic	2	0.20	0.40	No	Checkout desk	7A
SW1-SW2	Homing limit switch	KW12 microswitch	2	0.75	1.50	Yes	Amazon	7A
MECH1	Belt and pulley kit	GT2 6 mm belt 5 m + 2x 20T pulley + 4x idler	1	10.00	10.00	Yes	Amazon	7A
MECH2	Linear motion kit	4x 8 mm x 400 mm rod + 8x LM8UU bearing	1	18.00	18.00	Yes	Amazon	7A
MECH3	Gantry brackets and carriage	3D printed PLA	1	0.00	0.00	No	Texas Invention Works	7A
L1	Electromagnet	12 V 20 mm holding electromagnet (∼2.5 kg)	1	7.00	7.00	Yes	Amazon	7A
Q1	Electromagnet switch	Logic-level N-MOSFET (IRLZ44N / AO3400)	1	0.50	0.50	Yes	Amazon or checkout desk	7A
D1	Flyback diode	1N5819 Schottky	1	0.20	0.20	No	Checkout desk	7A
MAG1	Piece magnets	10 mm x 3 mm N52 neodymium (40 pack)	1	7.00	7.00	Yes	Amazon	7A
S1-S64	Hall sensor	TI DRV5032 omnipolar digital	64	0.45	28.80	No	TI free samples	7A
U4-U11	Shift register	TI SN74HC165	8	0.50	4.00	Yes	Digikey (or TI samples)	7A
PCB2	Sensor carrier boards	Perfboard or second PCB	1	0.00	0.00	No	TBD	7A
DS1	Display	1.8 in ST7735R TFT (Adafruit 358)	1	19.95	19.95	No	Already owned (earlier labs)	7A
SW3-SW5	Pushbutton	6 mm tactile switch	3	0.50	1.50	No	Checkout desk	7A
U12	SPI flash (stretch)	W25Q32 4 MB	1	0.60	–	Yes	Digikey	10
BRD1	Board top	2-3 mm acrylic or plywood sheet	1	6.00	6.00	Yes	Home Depot; laser cut at TIW	7A
PCS1	Chess pieces	3D printed PLA	32	0.00	0.00	No	Texas Invention Works	7A
PCB1	Main PCB	JLCPCB 2-layer ≤ 30 sq in	1	15.00	–	No	Ordered by professor	7B
-	Passives	Resistors and bypass caps	1	5.00	5.00	No	Checkout desk	7A
PWR1	12 V power supply	12 V 3 A barrel adapter	1	0.00	–	Yes	TBD	7B
PWR2	Regulators	12 V to 5 V buck + 3.3 V LDO	1	0.00	–	Yes	TBD	7B
ENC1	Enclosure	TBD	1	0.00	–	Yes	TBD	7B
				1a: all components	**151.35**			
				1b: counts toward limit	**79.00**			

Budget note. The 1b total of $79.00 is before the 12 V supply, regulators and enclosure. Ways to cut it: get the SN74HC165s as TI samples ($4), replace the rod-and-bearing kit with 3D-printed or V-slot rails (up to $18), and check the checkout desk for stepper motors ($20). If TI declines the Hall sensor samples, they add about $29 to the 1b total.